

# The Digital Comparator

Another common and very useful combinational logic circuit is that of the **Digital Comparator** circuit. Digital or Binary Comparators are made up from standard AND, NOR and NOT gates that compare the digital signals present at their input terminals and produce an output depending upon the condition of those inputs.

For example, along with being able to add and subtract binary numbers we need to be able to compare them and determine whether the value of input A is greater than, smaller than or equal to the value at input B etc. The digital comparator accomplishes this using several logic gates that operate on the principles of Boolean Algebra. There are two main types of Digital Comparator available and these are.

1. **Identity Comparator** – an *Identity Comparator* is a digital comparator that has only one output terminal for when  $A = B$  either “HIGH”  $A = B = 1$  or “LOW”  $A = B = 0$
2. **Magnitude Comparator** – a *Magnitude Comparator* is a type of digital comparator that has three output terminals, one each for equality,  $A = B$  greater than,  $A > B$  and less than  $A < B$

The purpose of a **Digital Comparator** is to compare a set of variables or unknown numbers, for example A ( $A_1, A_2, A_3, \dots A_n$ , etc) against that of a constant or unknown value such as B ( $B_1, B_2, B_3, \dots B_n$ , etc) and produce an output condition or flag depending upon the result of the comparison. For example, a magnitude comparator of two 1-bits, (A and B) inputs would produce the following three output conditions when compared to each other.

$$A > B, \quad A = B, \quad A < B$$

Which means: A is greater than B, A is equal to B, and A is less than B

## 1-bit Digital Magnitude Comparator:

### Truth Table:

| Inputs |   | Outputs |         |         |
|--------|---|---------|---------|---------|
| B      | A | $A > B$ | $A = B$ | $A < B$ |
| 0      | 0 | 0       | 1       | 0       |
| 0      | 1 | 1       | 0       | 0       |
| 1      | 0 | 0       | 0       | 1       |
| 1      | 1 | 0       | 1       | 0       |

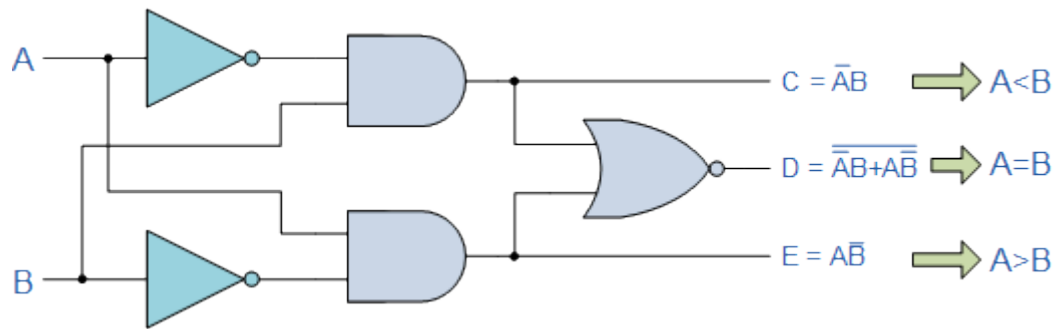
### Logic Expression:

$$X_0 = A_0 B_0 + A_0' B_0'$$

$$(A=B)=X_0, \quad (A<B)= A_0' B_0, \quad (A>B)= A_0 B_0'$$

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**Logic Diagram:**



### Figure 1: 1 Bit Comparator

### 2-bit Digital Magnitude Comparator:

### Logic Expression:

$$X_n = A_n B_n + A_n' B_n'$$

$$(A=B) = X_1X_0$$

$$(A < B) = A_1' B_1 + X_1 A_0' B_0$$

$$(A>B)=A_1B_1'+X_1A_0B_0'$$

### Block Diagram:

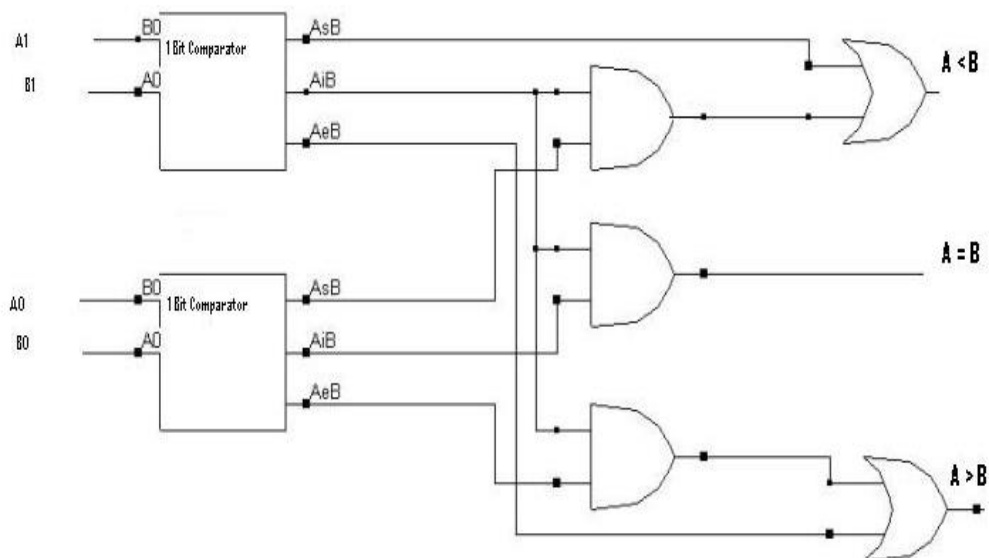


Figure 2: 2 Bit Comparator

### 3-bit Digital Magnitude Comparator:

#### Logic Expression:

$$X_n = A_n B_n + A_n' B_n'$$

$$(A=B) = X_2 X_1 X_0$$

$$(A < B) = A_2' B_2 + X_2 A_1' B_1 + X_2 X_1 A_0' B_0$$

$$(A > B) = A_2 B_2' + X_2 A_1 B_1' + X_2 X_1 A_0 B_0'$$

#### Block Diagram:

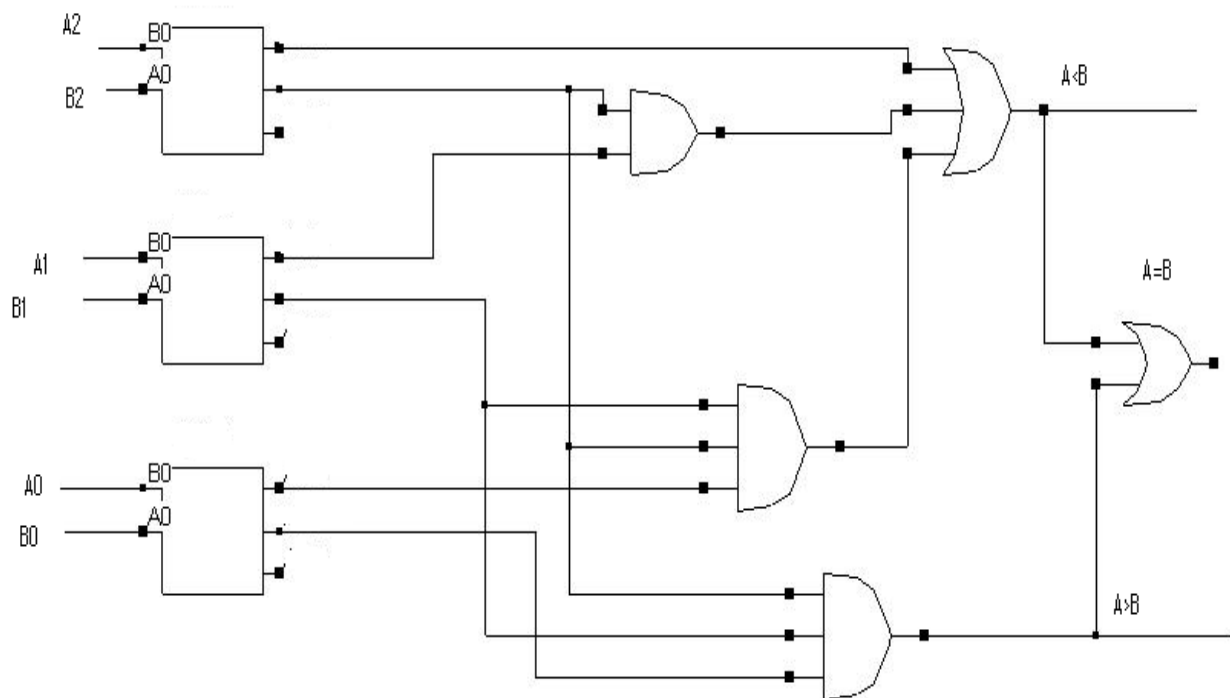


Figure 3: 3 Bit Comparator

### 5-bit Digital Magnitude Comparator:

#### Logic Expression:

$$X_n = A_n B_n + A_n' B_n'$$

$$(A=B) = X_4 X_3 X_2 X_1 X_0$$

$$(A < B) = A_4' B_4 + X_4 A_3' B_3 + X_4 X_3 A_2' B_2 + X_4 X_3 X_2 A_1' B_1 + X_4 X_3 X_2 X_1 A_0' B_0$$

$$(A > B) = A_4 B_4' + X_4 A_3 B_3' + X_4 X_3 A_2 B_2' + X_4 X_3 X_2 A_1 B_1' + X_4 X_3 X_2 X_1 A_0 B_0'$$

## Block Diagram:

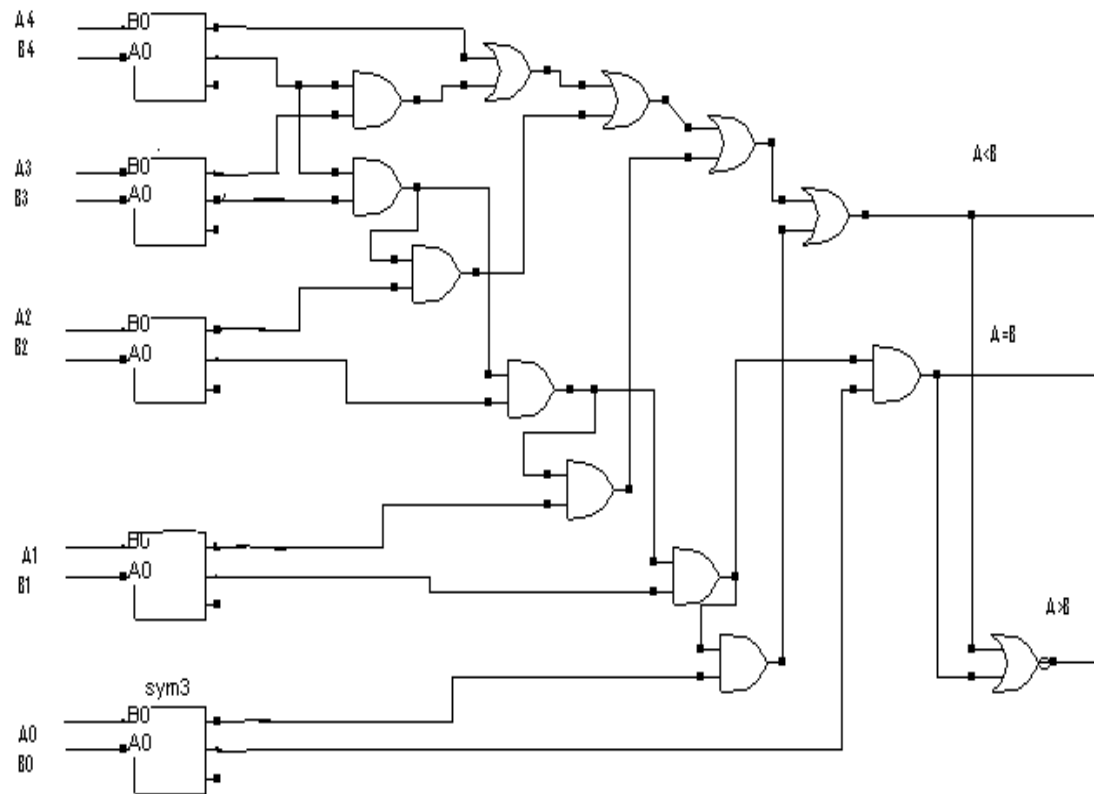


Figure 4: 5 bit comparator